II. Adventitious root modification

a. Storage roots

1. Tuberous root

These roots are swollen without any definite shape. Tuberous roots are produced singly and not in clusters. Example: *Ipomoea batatas.*

2. Fasciculated root

These roots are in cluster from the base of the stem Example: *Dahlia, Asparagus.*

3. Nodulose root

In this type of roots, swelling occurs only near the tips. Example: *Maranta* (Arrow root) *Curcuma amada* (Mango ginger), *Curcuma longa* (Turmeric)

4. Moniliform or Beaded root

These roots swell at frequent intervals giving them a beaded appearance. Example: *Vitis, Portulaca, Momordica.*

5. Annulated root

These roots have a series of ring- like swelling on their surface at regular intervals. Example: *Psychotria* (Ipecac)

b. Mechanical support

1. Prop (Pillar) root

These roots grow vertically downward from the lateral branches into the soil. Example: *Ficus benghalensis* (banyan tree), Indian rubber.

2. Stilt (Brace) root

These are thick roots growing obliquely from the basal nodes of the main stem. These provide mechanical support. Example: *Saccharum officinarum, Zea mays, Pandanus and Rhizophora*.

3. Climbing (clinging) roots

These roots are produced from the nodes of the stem which attach themselves to the support and help in climbing. To ensure a foothold on the support they secrete a sticky juice which dries up in air, attaching the roots to the support. Example: *Piper betel.*

4. Buttress root

In certain trees broad plank like outgrowths develop towards the base all around the trunk. They grow obliquely downwards and give support to huge trunks of trees. This is an adaptation for tall rain forest trees. Example: *Bombax ceiba* (Red silk cotton tree), *Ceiba pentandra* (white silk cotton tree), *Delonix regia, Bombax.*



Ipomoea batatas



Dahlia



Maranta



Psychotria

Figure 3.5: Adventitious Root Modification for Storage



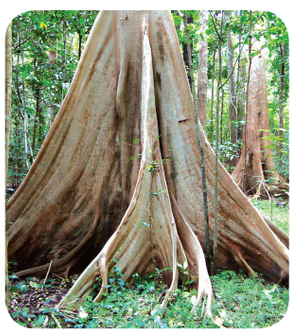
Ficus benghalensis



Saccharum officinarum



Epipremnum pinnatum



Bombax

Figure 3.6: Adventitious root modifications for mechanical support

XIth_Botany_Chapter_3_EM.indd 65 5/17/2020 9:02:33 PM